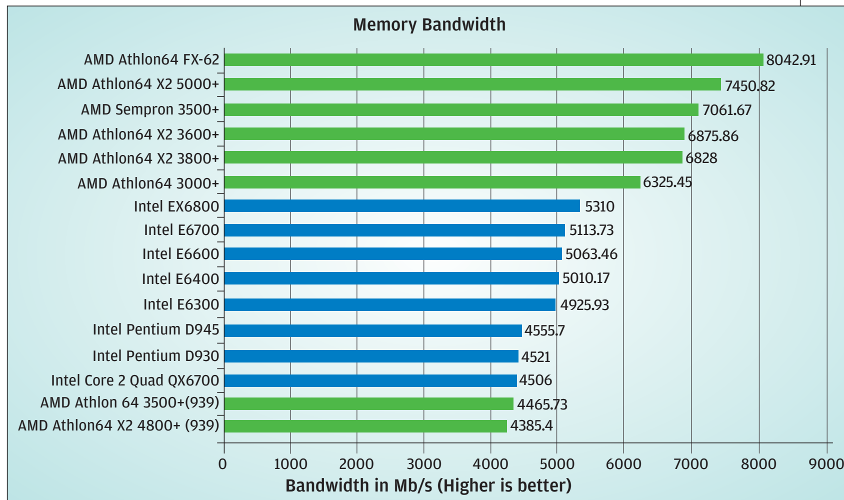
ScienceMark 2.0 Memory Bandwidth

Surprised? Don't be! The onboard memory controller design in AMD's K8 range of processors is very efficient and offers huge bandwidth. In fact, even the Athlon64 AM2 3000+ offers a higher bandwidth than the EX6800. What we are talking about here is the available bandwidth and how effectively it is utilised. The Quad Core is way down on the list, and that might be due to the overheads introduced by the interface between processors. The Quad Core is actually two dual cores slapped on to become one, which isn't really an ideal design.



How we Tested

We chose benchmarks that would stress the processor to the hilt. We also made sure that the benchmarks had a low level of sub-system involvement to obtain true processor performance.

All Intel LGA 775-based processors were plugged into Intel's D975XBX motherboard with 2 GB of Corsair 800 MHz memory with 4-4-4-12 timing figures. For the AMD AM2 platform, we used ASUS' Crosshair motherboard with 2 GB of Corsair 800 MHz memory. For the 939-pin processor, we used ASUS' A8R32-MPV deluxe with 2 GB of Corsair DDR 400 MHz memory.

Windows XP with SP2 was freshly loaded on a Seagate 7200 RPM SATA drive. XFX 7900GTX card was the standard graphics card used with both platforms.

Gaming and Multimedia

PCMark 05, 3DMark05 and 3DMark 06

These benchmarks from FutureMark are widely used to gauge the performance of a system, on a system as well as sub-system level. Though mostly related to graphics performance, each of these benchmarks has a well-developed CPU test that taxes the processor to its maximum. We used the default setting and chose only the CPU test. Each test was run three times before noting down the scores.

Far Cry and Doom 3

Both these games boast of high graphical details and are quite taxing on the processor making them ideal processor benchmarks. We ran both at 640x 480, 800 x600 and 1024 x 768 resolutions to put the maximum possible stress on CPU and not the GPU—at lower resolutions, games offload more tasks on to the CPU.

Ziff Davis Multimedia Content Creation and Business Winstone

Multimedia Content Creation and Business Winstone run applications such as Adobe Photoshop, Adobe Premiere, WaveLab, anti-virus, compression, word-processing, etc. The result is a unified score that reflects the performance of the system as a whole.

DivX Encoding

We encoded a 100 MB VOB file to DivX using the DivX converter tool. The time taken to encode the file was noted down with each processor. The benchmark was run thrice for each processor.

3D rendering benchmarks

Cinebench 9.2

Cinebench 2003 is the free benchmarking tool for Windows and Mac OS based on the 3D software Cinema 4D engine. Cinebench 2003 delivers accurate results by testing the processors' raw processing speed with OpenGL, multithreading and multiprocessing. This benchmark was used to indicate the performance gain due to multi-processing.

POV-Ray 3.6

The Persistence Of Vision Raytracer, a tool that allows creating an image by mathematical calculations. The number-crunching involved is purely processor-intensive, which is why it works as a great processor benchmark. In the default installation we used the 'Chess2' and 'landscape' to be rendered at 1024 x768 resolution without AA.

Kirbi Bench 1.1

Another rendering benchmark, based on the Kirbi engine developed by Adept software. This tool also consists of many models with multiple light sources, and the rendering is done real time. The score is given in terms of frames the processor is able to churn out.

Number-crunching Tests

ScienceMark 2.0

ScienceMark 2.0, as the name suggests, is a benchmark based on scientific calculations. It consists of multiple benchmarks that test the various aspects of a processor. We used the MolDyn, Primordia and Cipher test.

WinRAR 3.6

WinRAR 3.6 offers an inbuilt benchmark, operating on some random data which is processed in the memory. Since the data is operated on in the memory, the results are free from any influences that the hard disk's performance may have on it.

SiSoft Sandra 2007

This is the good old benchmarking tool with new features and tests added to it. We logged scores particularly related to the CPU—Dhrystone, Whetstone, Multimedia Index (Integer and Float) and Cache performance with a 4 MB data block.

Super Pi

A small utility that calculates the value of pi up to 32 million places. We used the utility to calculate the value of pi, starting from 120,000 places right up to 8 million. The time taken to complete the calculation was noted down in each case.